

Linear Algebra — Exercises

Complex numbers

Exercise 1. Calculate: a) $(2+3i)(4-2i)$ b) $(1+2i)(2i-1)$ c) $(3+i)(3-i)$ d) $(a+bi)(a-bi)$

Exercise 2. Calculate: a) $\frac{2-3i}{3+4i}$ b) $\frac{i}{1+i}$ c) $\frac{3-4i}{i}$

Exercise 3. Calculate: a) $\frac{1+4i}{4-i}$ b) $\frac{(3+4i)^4}{(3-4i)^3}$

Exercise 4. Show that for every two complex numbers z and w :

a) $\overline{z+w} = \bar{z} + \bar{w}$ and $\overline{zw} = \bar{z} \bar{w}$

b) $|zw| = |z| |w|$

c) $\left| \frac{z}{w} \right| = \frac{|z|}{|w|}$

Exercise 5. Show that multiplication of complex numbers is associative.

Exercise 6. Show that multiplication of complex numbers is distributive with respect to addition.

Exercise 7. Describe geometrically the sets of points $z \in \mathbb{C}$ satisfying the following conditions:

a) $|z - i + 3| = 5$

e) $\Re((z-i)^2) = 0$

b) $|z - i + 3| > 5$

f) $\Re(z^2) = 2$ and $(\Im(z+i))^2 = 1$

c) $|z - 1 + 3i| \leq 5$

g) $\left| \frac{z+3}{z-2i} \right| \geq 1$

d) $2 \leq |z+i| < 4$

h) $\Re(z+1) < 0$ and $|i-z| \leq 3$

Exercise 8. State which of the following are groups: a) $(\{z \in \mathbb{C} \mid |z| = 1\}, +)$

b) $(\{z \in \mathbb{C} \mid |z| = 1\}, \cdot)$

Exercise 9. Find polar forms of the following complex numbers:

a) $1+i$

g) $1-i\sqrt{3}$

b) -3 (negative 3)

h) $-z$, if $z = \cos \alpha + i \sin \alpha$

c) $-i$

i) $\sqrt{3} - i$

d) $2+2i$

j) $\bar{z}$, if $z = \cos \alpha + i \sin \alpha$

e) $3+3\sqrt{3}i$

k) $\frac{1}{2} + \frac{\sqrt{3}}{2}i$

f) $z = \sin \alpha + i \cos \alpha$

Exercise 10. Calculate:

a) $\sqrt[3]{(1+i)^6}$

d) $\left(\frac{2+2i}{\sqrt{3}-i} \right)^8$

b) $\left(\frac{1-i}{1+i} \right)^{222}$

e) $\sqrt[6]{1}$

c) $\left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)^{124}$

f) $\sqrt[4]{-1}$

g) $\sqrt[4]{(3-5i)^4}$

h) $\sqrt[6]{-1}$

Exercise 11. Describe geometrically the sets of all the following roots: a) $\sqrt[6]{1}$ b) $\sqrt[4]{-1}$ c) $\sqrt[8]{1}$
d) $\sqrt[6]{-1}$

Exercise 12. Show that for every n all complex roots of 1 of order n form a group under multiplication. Write down the table of the group of all complex roots of 1 of order 4.

Exercise 13. Use De Moivre's formula to:

- (a) express $\sin 5\alpha$ in terms of $\sin \alpha$ and $\cos \alpha$,
- (b) express $\cos 4\alpha$ in terms of $\sin \alpha$ and $\cos \alpha$.

Exercise 14. Prove that

$$\frac{\cos \alpha + i \sin \alpha}{\cos \beta + i \sin \beta} = \cos(\alpha - \beta) + i \sin(\alpha - \beta).$$

Exercise 15. Find the polar form of: a) $2 + \sqrt{2} + i\sqrt{2}$ b) $2 + i + i\sqrt{3}$

Exercise 16. Calculate: a) $\cos \frac{\pi}{12}$ b) $\cos 105^\circ$ c) $\cos 75^\circ$ d) $\cos(-15^\circ)$

Exercise 17. Calculate $(1 + \cos \alpha + i \sin \alpha)^{10}$ for $\alpha \in [0, \pi)$.

Exercise 18. Let $f(x) \in \mathbb{R}[x]$. Show that z is a root of f if and only if $\bar{z}$ is a root of f .

Exercise 19. Factor each of the following polynomials into a product of factors of degree at most 2:
a) $x^6 + 1$ b) $x^4 + 1$ c) $x^6 + x^3 + 1$ d) $x^5 + 1$

Exercise 20. Prove that every polynomial of an odd degree, with all real coefficients, has at least one real root.

Exercise 21. Knowing that $1 - 2i$ is a root of the polynomial $z^4 + 2z^3 + 2z^2 + 10z + 25$, find all remaining roots.

Exercise 22. Knowing that $2 + i$ is a root of the polynomial $z^4 - 2z^3 - 6z^2 + 22z - 15$, find all remaining roots.

Exercise 23. Solve the equations: a) $x^2 + x + 1 = 0$ b) $2x^2 + ix + 3 = 0$ c) $x^2 + 2x - i\sqrt{3} = 0$